

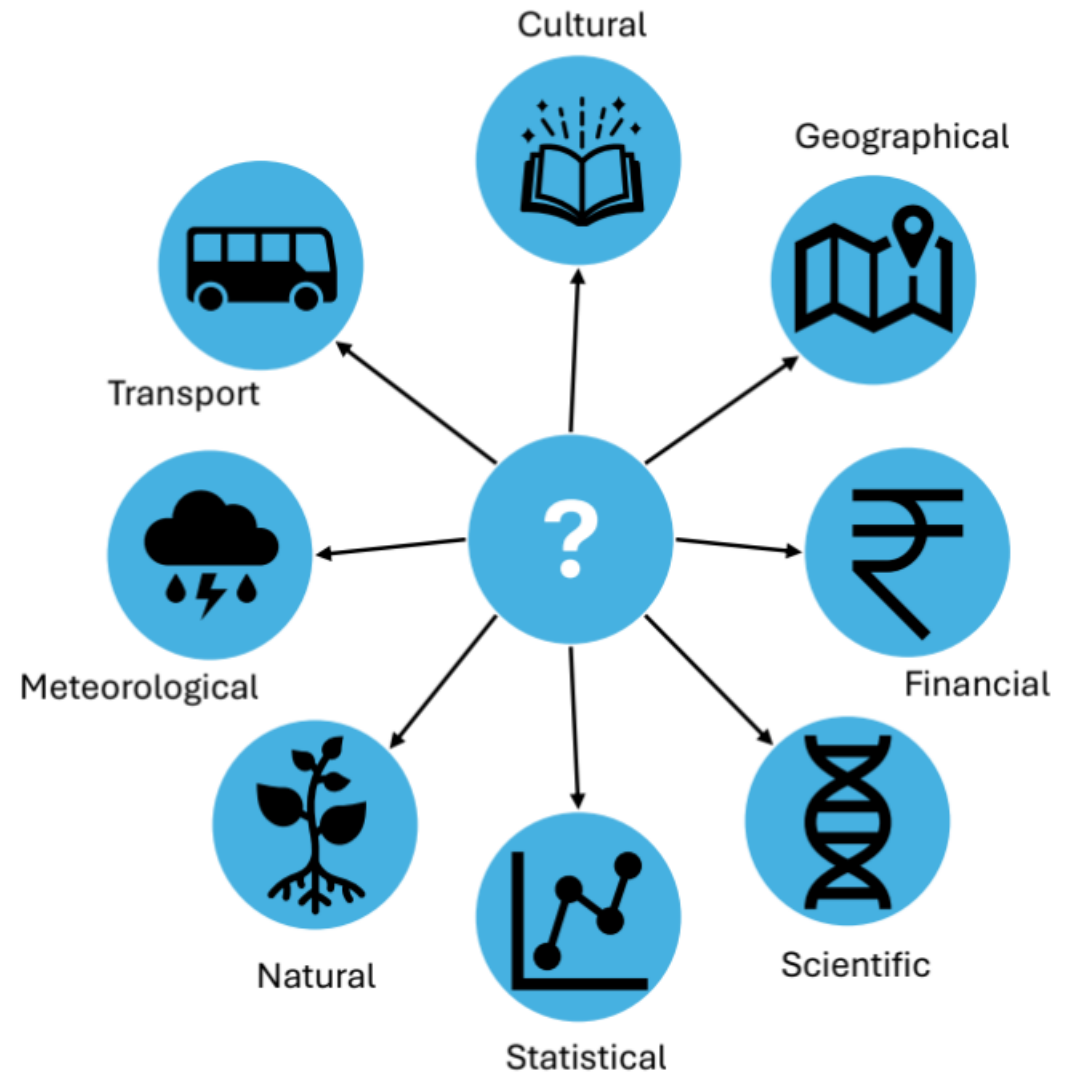
Structured Data

Software System Development
Monsoon 2026

What is Data?

- Data is the foundational substrate of all Software Systems.
- Without data, systems are merely stateless logic engines waiting for input.
- Data provides the state, memory, and context that give software its purpose.

**If 'Software System' is BODY
Then 'Data' is its SOUL!**



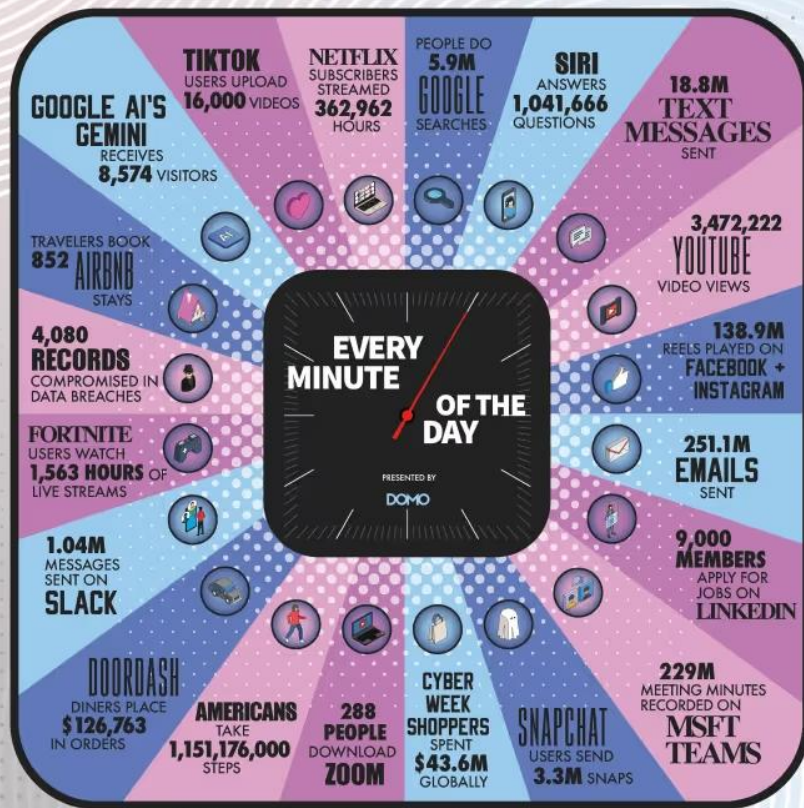
DATA NEVER SLEEPS



Every minute of every day, the world generates a dizzying amount of data, and how we interact with it is constantly changing. AI tools are now answering millions of questions in real time and are transforming how we work, shop, and connect. Digital platforms are seeing explosive usage, with billions of emails, texts, and reels shared every day. Entertainment continues to drive engagement across streaming, gaming, and social media while e-commerce is setting new benchmarks as digital habits evolve and expand at an unprecedented pace.

In Domo's 12th edition of Data Never Sleeps, we capture a snapshot of this world powered by the rapid rise of data, AI, and digital activity, shaping every moment of modern life.

12



The world's internet population continues to grow significantly year-over-year. As of late 2024, 5.52 billion people—approximately 67.5% of the global population—are online.

According to industry analysts, the total amount of data created, captured, copied, and consumed globally is expected to reach 149 zettabytes by the end of 2024, with projections surpassing 394 zettabytes by 2028.

Global Internet Population Growth
(IN BILLIONS)



As the volume and complexity of data accelerates, business success increasingly depends on the ability to turn information into insights. Domo helps you harness the power of data and AI so you can adapt as quickly as the world changes and make data-driven decisions that set you apart. Let Domo help you make sense of all the clicks, swipes, and streams so you can see the big picture shaped by every small decision.

Learn more at domo.com

SOURCES: EARTHWEB, DUSTIN STOLT, DEMANDSAGE, HOOTSUITE, BUSINESSWIRE, DOORDASH, SOCIALPLOT, X | TWITTER.COM, GITHUB, INVGATE, THINKIMPACT, SPINA.DIGI, STATISTA, PR NEWS-WIRE, NETSCOUT



www.data.gov.in

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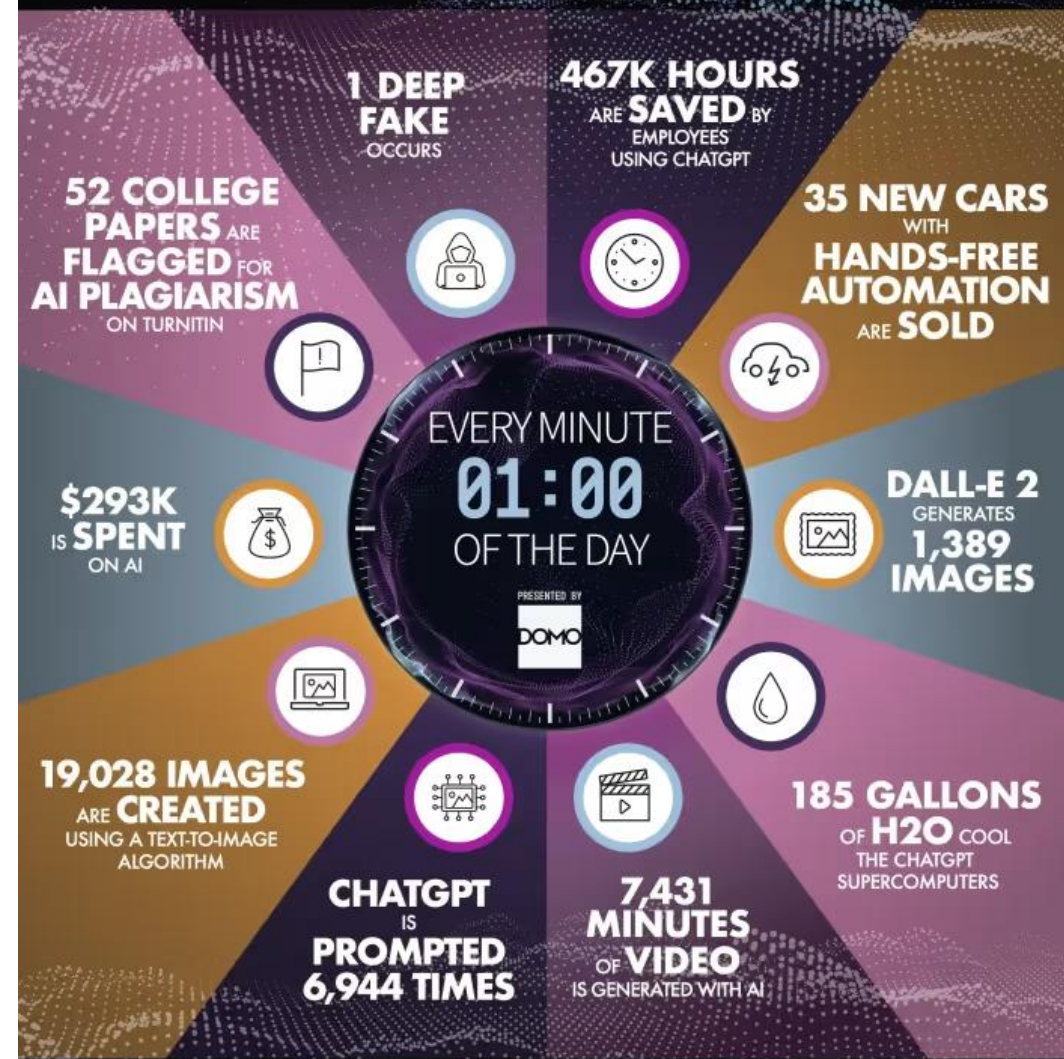


DATA NEVER SLEEPS AI EDITION

Domo's been monitoring the world's data usage – by the minute – for over a decade now, and the AI phenomenon has caught us all by storm. This recent AI boom has transformed the way businesses operate, students learn and the way people live their everyday lives.

In this special edition of Data Never Sleeps, focused on AI, we dive deeper into the immense amount of data generated by AI every minute and its impact on our lives. Numbers speak volumes, and we're witnessing just the beginning of a new and powerful era of AI.

Learn more at Domo.AI

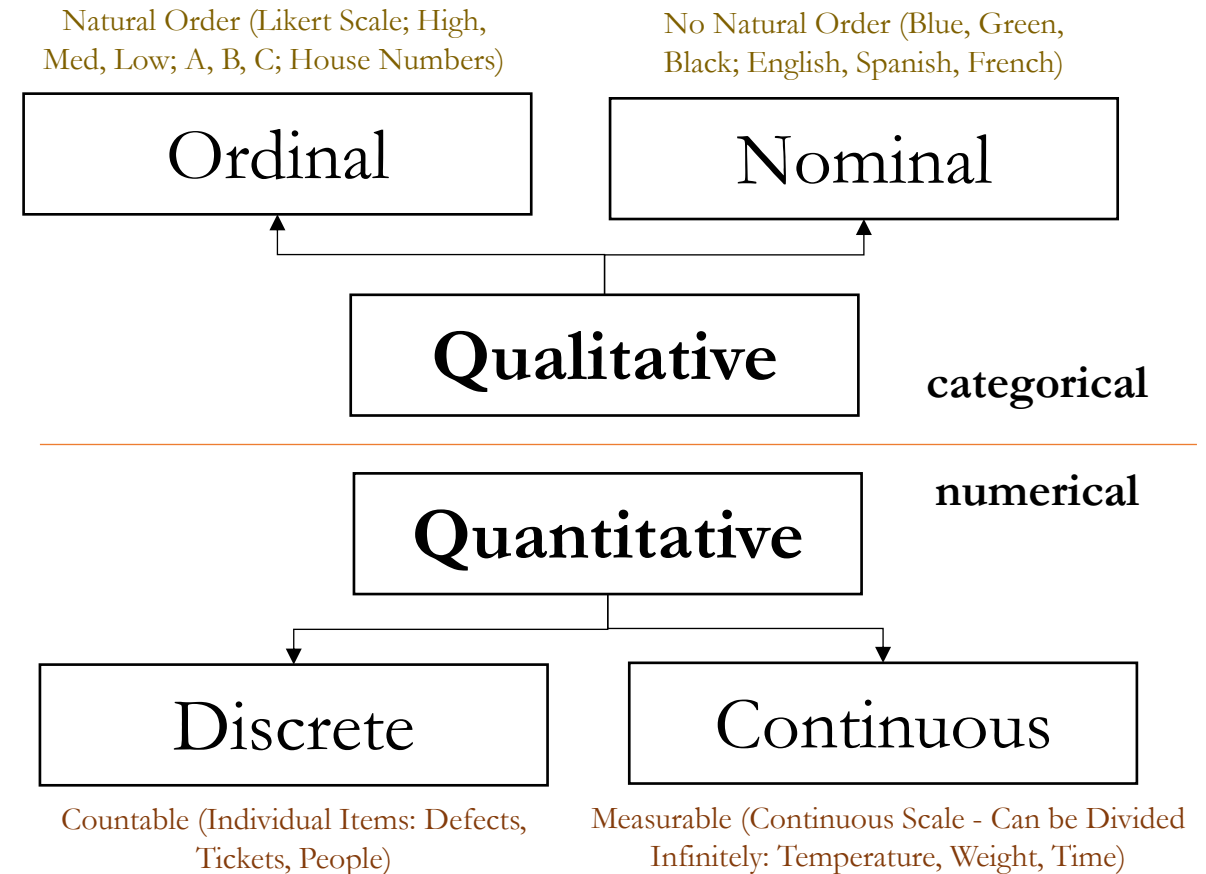


SOURCES: AnalyticsIndiaMag, Invgate, TheHindu, MarketingProfs, Niteshbora.com, Reuters, Statista

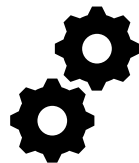
Data Explosion

Much of this data is now stored in traditional Relational

A major challenge is to manage this data, answer queries over it, gather interesting and useful insights from it.



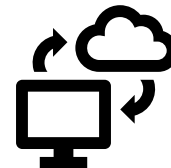
DATA → **KNOWLEDGE** → **ACTION**



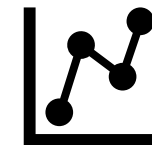
organize



store

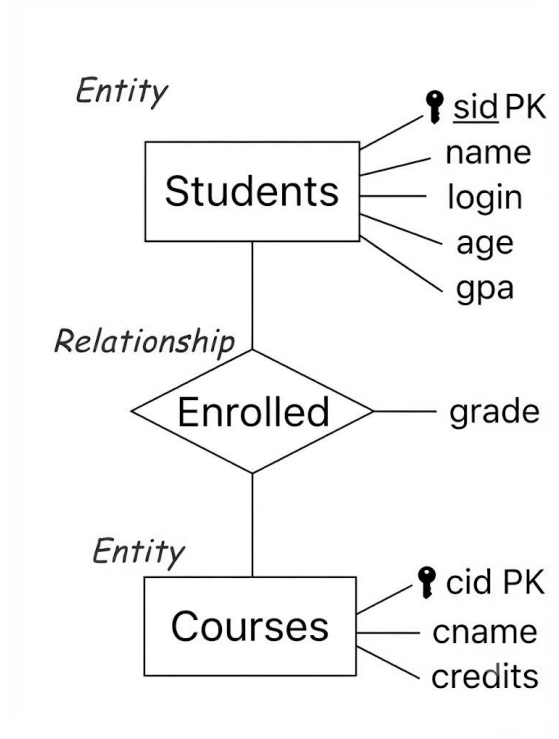


share



analyze

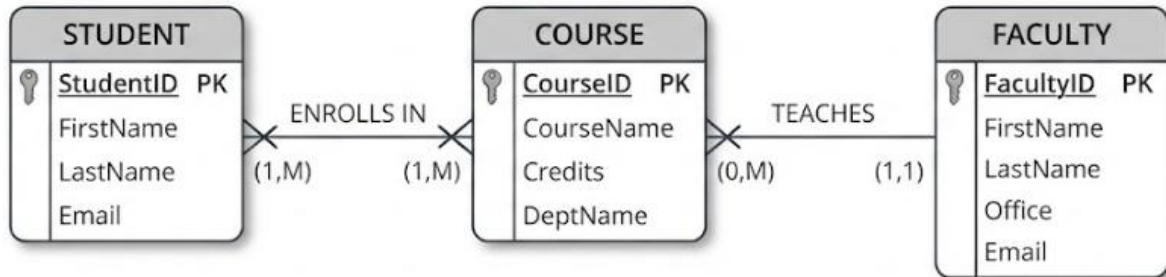
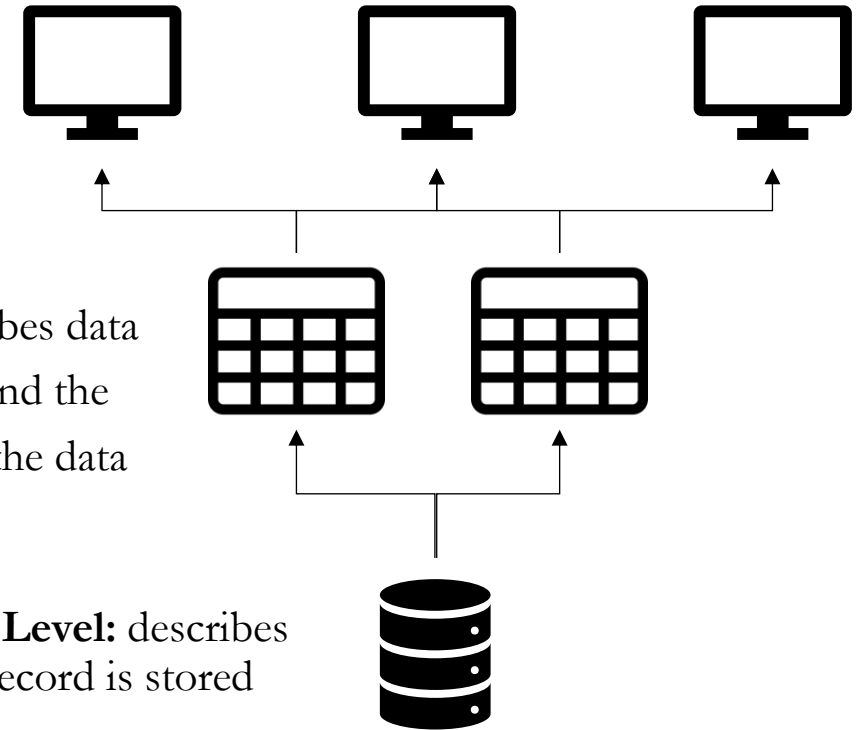
Data Abstraction Levels



View level: application programs hide details of data types. Views can also hide information (such as an employee's salary) for security purposes.

Logical Level: describes data stored in database, and the relationships among the data

Physical Level: describes how a record is stored



**Levels of Data Abstraction
in Software Systems**

Data Independence

Logical data independence: Protection from changes in logical structure of data.

If a user application operates on a subset of attributes of a relation, it should not affect when new attributes are added. Can make changes without affecting existing schemas.

Physical data independence: Protection from changes in physical structure of data.

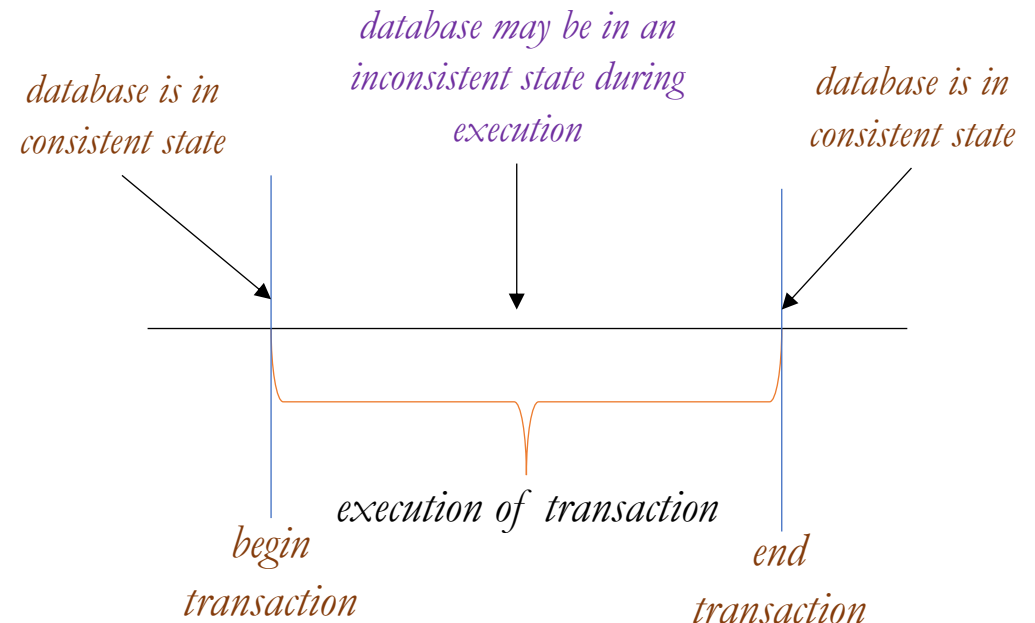
Change in internal data using different file organization or devices should be possible without changing logical schema structure.

Atomicity: If any operation is performed on the data, either it should be performed or executed completely or should not be executed at all.

Consistency: Integrity of the data should be maintained, which means if a change in the database is made, it should remain preserved always

Isolation: Transactions occur independently without interference. Changes should be visible only after they have been made to the main memory

Durability: Data after the successful execution of the operation becomes permanent in the database.



Data Definition Language – Define and Manage

CREATE, ALTER, DROP, TRUNCATE, RENAME, COMMENT

Data Manipulation Language – Manipulates data

SELECT, INSERT, UPDATE, DELETE, MERGE, CALL, EXPLAIN PLAN, LOCK TABLE

REDUNDANCY

Insertion anomalies: It may be impossible to store certain information without storing some other, unrelated information.

Deletion anomalies: It may be impossible to delete certain information without losing some other, unrelated information.

Update anomalies: If one copy of such repeated data is updated, all copies need to be updated to prevent inconsistency.

NORMALIZATION

It is the process of structuring a relational database in accordance with a series of normal forms in order to **reduce data redundancy** and **improve data integrity**.

Normal Form	Core Requirement	Example
First Normal Form (1NF)	Atomic values only (no repeating groups).	Violation: A cell containing "Python, SQL". Fix: Separate into individual rows or a child table.
Second Normal Form (2NF)	No partial dependencies on a composite primary key.	Violation: Key is (StudentID, CourseID), but InstructorName depends only on CourseID. Fix: Move InstructorName to a separate Courses table.
Third Normal Form (3NF)	No transitive dependencies (non-key depending on non-key).	Violation: EmployeeID → DepartmentID → DepartmentName. Fix: Move DepartmentName to a separate Departments table.
Boyce-Codd Normal Form (BCNF)	Every determinant must be a super key.	Violation: Advisor determines Course, but Advisor is not a candidate key. Fix: Split into (Advisor, Course) and (Student, Advisor) tables.
Fourth Normal Form (4NF)	No independent multi-valued dependencies.	Violation: An employee row repeats for every combination of independent Skill and Language lists. Fix: Split into separate EmployeeSkills and EmployeeLanguages tables.
Fifth Normal Form (5NF)	Lossless decomposition of cyclic dependencies.	Violation: A 3-way relationship (Agent, Supplier, Product) that can be incorrectly reconstructed if split naively. Fix: Decompose into precise binary tables that guarantee exact reconstruction.